PANDITA AI

AI for Data Centres

More capacity. Lower costs. Reliable operations.

Live Platform: Aether-DCTWIN • Hybrid Air + Direct-to-Chip Liquid • 100% Facility-Side (Zero Tenant Agents)

Practical AI and live facility digital twin for your next stage of growth

Updated Product & Use-Case Overview

What Pandita does

We bring AI research and engineering together to solve business problems.

Connect your data

Bring facility meters, equipment trends, engineering drawings and business documents together.

Build custom models

Combine physics guardrails and tailored AI models for your site’s exact cooling and power layout.

Automate workflows

Turn insights into safe setpoint recommendations and checklists, with engineers approving key decisions.

Backed by Evercloud AI, the AI arm of Everstone

Enterprise engagements: Vodafone • Constl • Tata Communications

AI is reshaping the data centre

AI is driving high-density demand. It can also help operators unlock more from existing sites.

~2×

Global data centre electricity demand by 2030

Compared with 2025
IEA forecast, 2026

Power limits growth

Reduce over-cooling waste and reclaim stranded kW for revenue-generating racks.

Air and liquid cooling now run side by side

Manage 15 kW perimeter-air rows alongside 125 kW+ direct-to-chip liquid AI pods safely.

Complexity slows decisions

Connect live telemetry, thermal physics and procedures so engineers act faster.

DIRECTION OF TRAVEL

Understand conditions → **Anticipate problems** → **Act with approval**

THE OPPORTUNITY

More from your data centre

Four operational priorities — now demonstrated live in Pandita's facility digital twin.

01 COOLING

Lower energy costs

Continuously tune cooling settings and peak-tariff battery shaving as load and weather change.

02 CAPACITY

More usable IT capacity

Simulate row expansions and 10,000-GPU liquid buildouts on existing and planned power.

03 MAINTENANCE

Earlier warning of problems

Catch strainer fouling and heat-exchanger drift early, with ready-to-run isolation checklists.

04 FAULT RESPONSE

Faster fault resolution

Pinpoint root-cause alarms on a 3D floor, inspect 42U racks and rewind 24 hours of history.

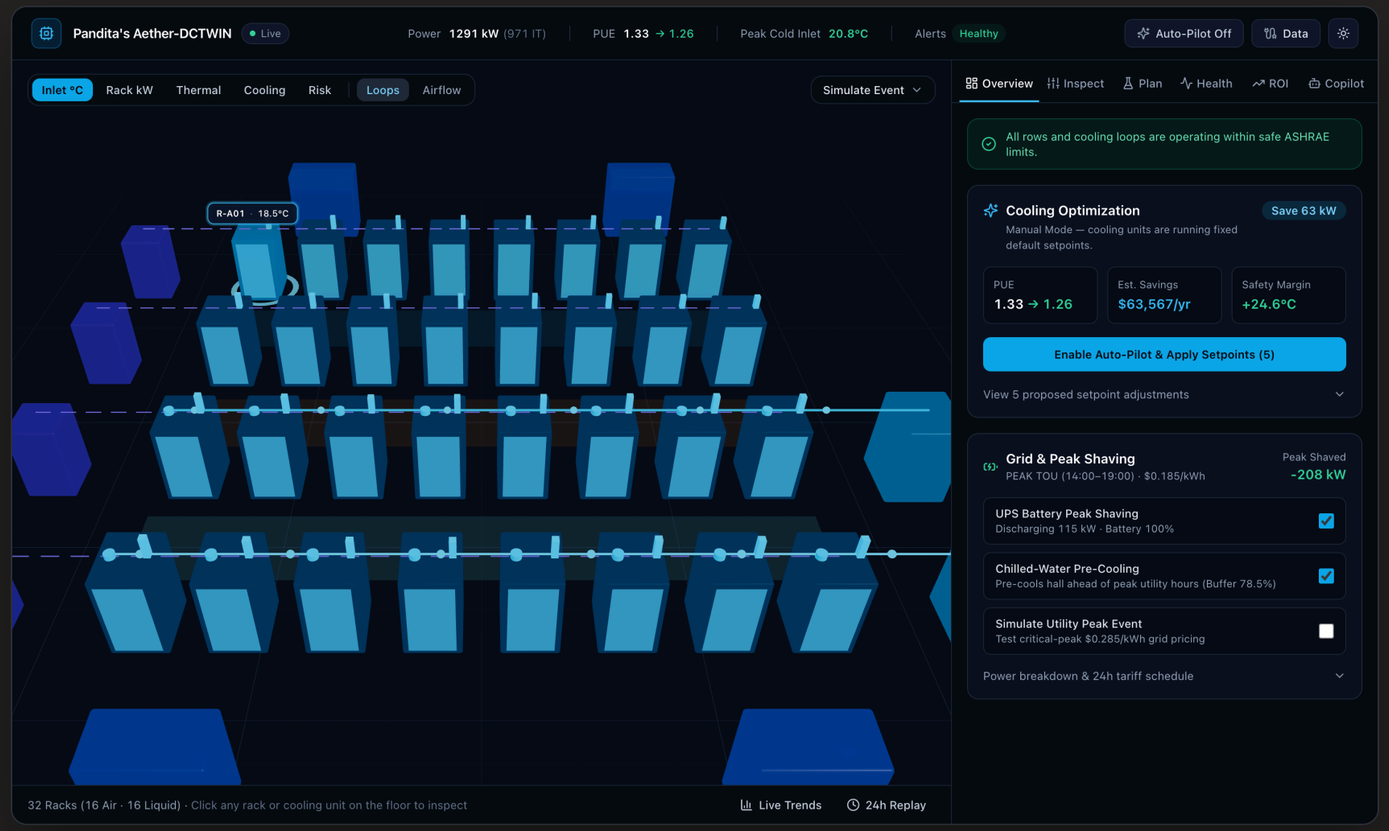
PLUS ENTERPRISE BUSINESS PRODUCTIVITY

Faster customer responses. Less operational admin.

Marketing **Sales** **Proposals** **Procurement** **Finance** **HR**

Pandita’s Aether-DCTWIN in action

An uncluttered, real-time 3D digital twin and decision workspace built for critical facility teams.



LIVE SCREENSHOT • OVERVIEW

32-rack hybrid hall (16 Air + 16 Liquid) proposing -63 kW savings (PUE 1.33 → 1.26)

100% facility-side boundary

Connects to BMS, DCIM and PDU meters via read-only BACnet, Modbus, SNMP or CSV. Zero software agents ever touch customer servers.

Clean, calm operator experience

Replaces cluttered legacy BMS screens with a clean 3D floor view, plain-English alerts and six task-focused workspaces.

Built-in safety & instant fallback

Every recommendation is checked against ASHRAE thermal limits and dew-point margins. One click reverts setpoints to fixed manual defaults.

Use less energy for cooling

Continuously match chillers, CRAC fans and CDU pumps to real IT heat load and utility pricing.

Eliminate over-cooling waste

Raises cold-aisle supply from fixed 18.6°C manual defaults to safe 22–23°C targets, cutting ~66 kW of fan and chiller draw (PUE 1.33 → 1.26).

Coordinate peak-tariff shaving

Pre-cools chilled-water thermal buffers ahead of peak utility hours and dispatches UPS battery peak shaving (-207 kW grid draw).

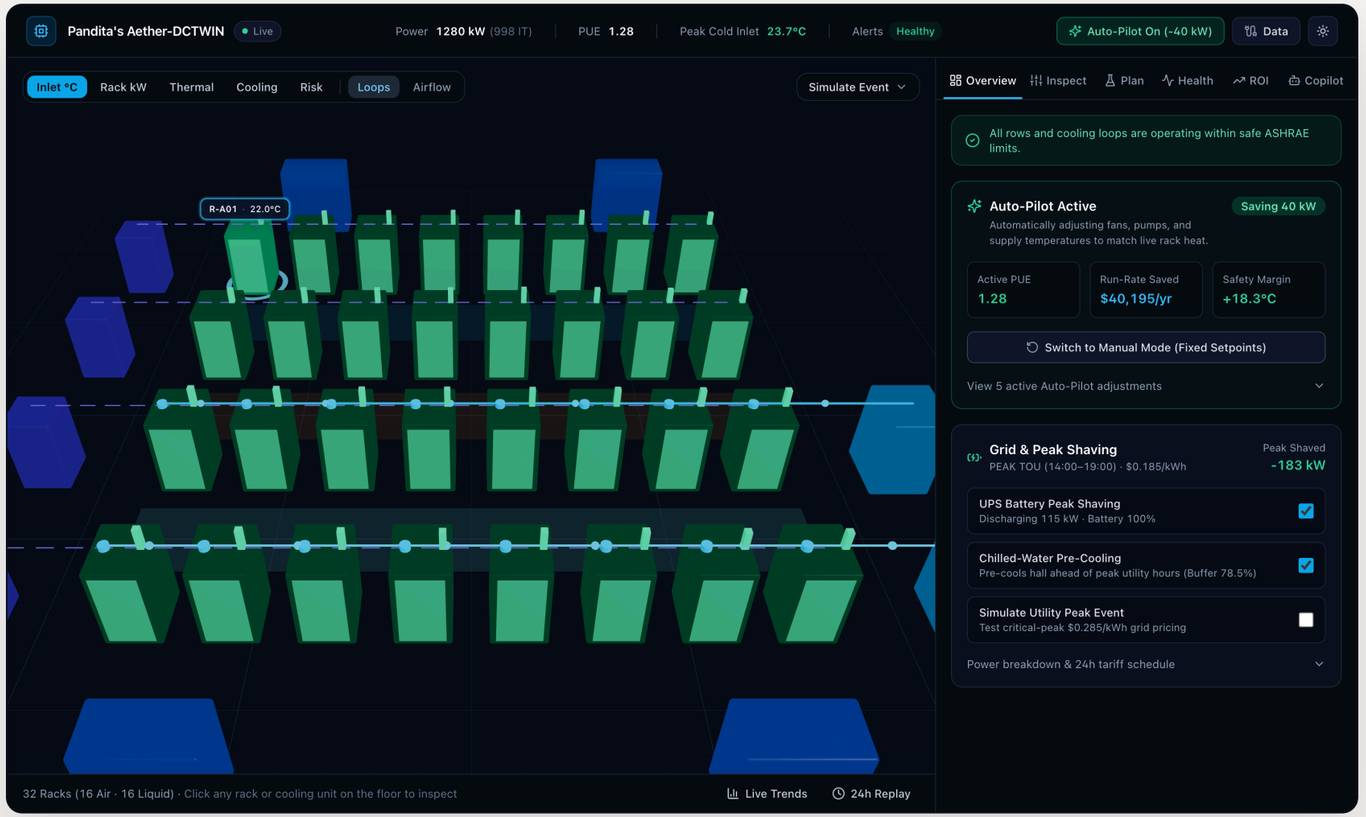
Approve once or run Auto-Pilot

Engineers inspect exact proposed setpoint changes before enabling Auto-Pilot, with continuous safety margin tracking.

INDUSTRY BENCHMARK • EQUINIX

900 MWh/yr

Energy reduction reported at its FR6 site after AI cooling optimisation.



LIVE PLATFORM • AUTO-PILOT & PEAK SHAVING

Real-time PUE drop (1.33 → 1.26) and -207 kW peak tariff shaving

Get more from existing power & plan AI scale-out

Test single-row GPU expansions today and engineer multi-hall 10,000-GPU training clusters with physics checks.

Row-level What-If sandbox

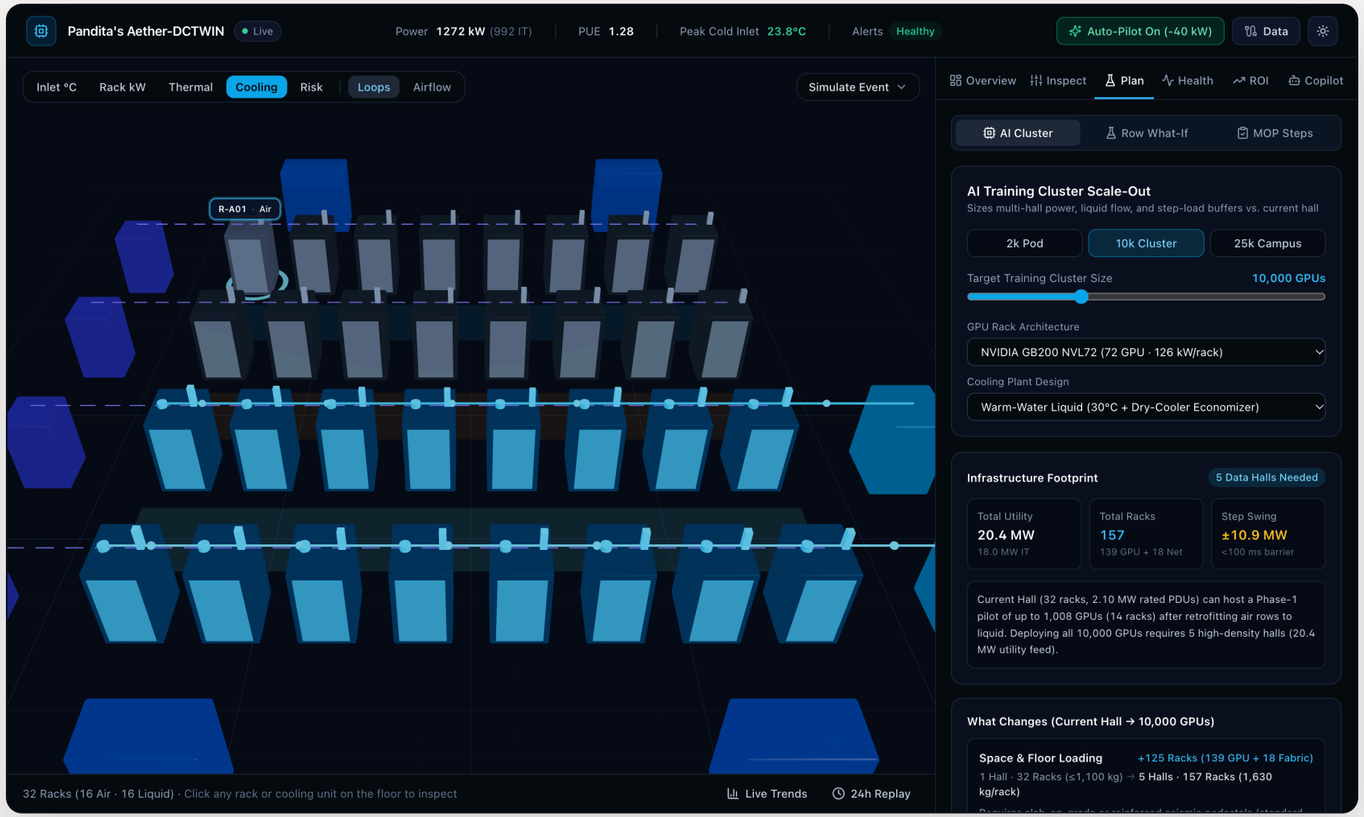
Test adding +160 kW of liquid GPUs to an existing row and verify thermal, CDU flow and breaker margins before installing hardware.

10,000-GPU cluster capacity planner

Models GB200 NVL72 & HGX H200 buildouts: sizes 5 data halls, 20.4 MW utility power, 157 racks (1,630 kg/rack slab load) and 21,400 LPM liquid flow.

Catch physical bottlenecks early

Flags 415V 3-phase busway upgrades, blocks infeasible air-only cooling (>35 kW/rack) and sizes <100 ms BESS buffers for ±10.9 MW training load swings.



LIVE PLATFORM • PLAN → AI CLUSTER

10k-GPU GB200 NVL72 architectural sizing alongside hybrid Air/Liquid 3D hall

Spot equipment problems earlier

Detect gradual mechanical fouling from existing meters before a degraded unit triggers a thermal incident.

Catch strainer & heat-exchanger drift

Tracks pump pressure drop, approach temperature and zone calibration to flag early CDU strainer restriction while temperatures still look normal.

30-minute thermal lookahead

Projects rack inlet and liquid temperatures 5, 15 and 30 minutes ahead so shift leads know exact thermal runway before limits are reached.

Automated isolation checklists (MOPs)

Generates step-by-step maintenance procedures mapped to redundant cooling loops and affected downstream racks.

INDUSTRY BENCHMARK • COMPASS DATACENTERS

40% fewer

On-site maintenance interventions, as reported by Schneider Electric.



LIVE PLATFORM • HEALTH WORKSPACE

CDU-1 fouling drift detection and 30-minute thermal safety forecast

Find faults faster

Cut through cascading rack alarms by linking floor-wide symptoms directly to the failed cooling or power asset.

Group cascading alarms into one root cause

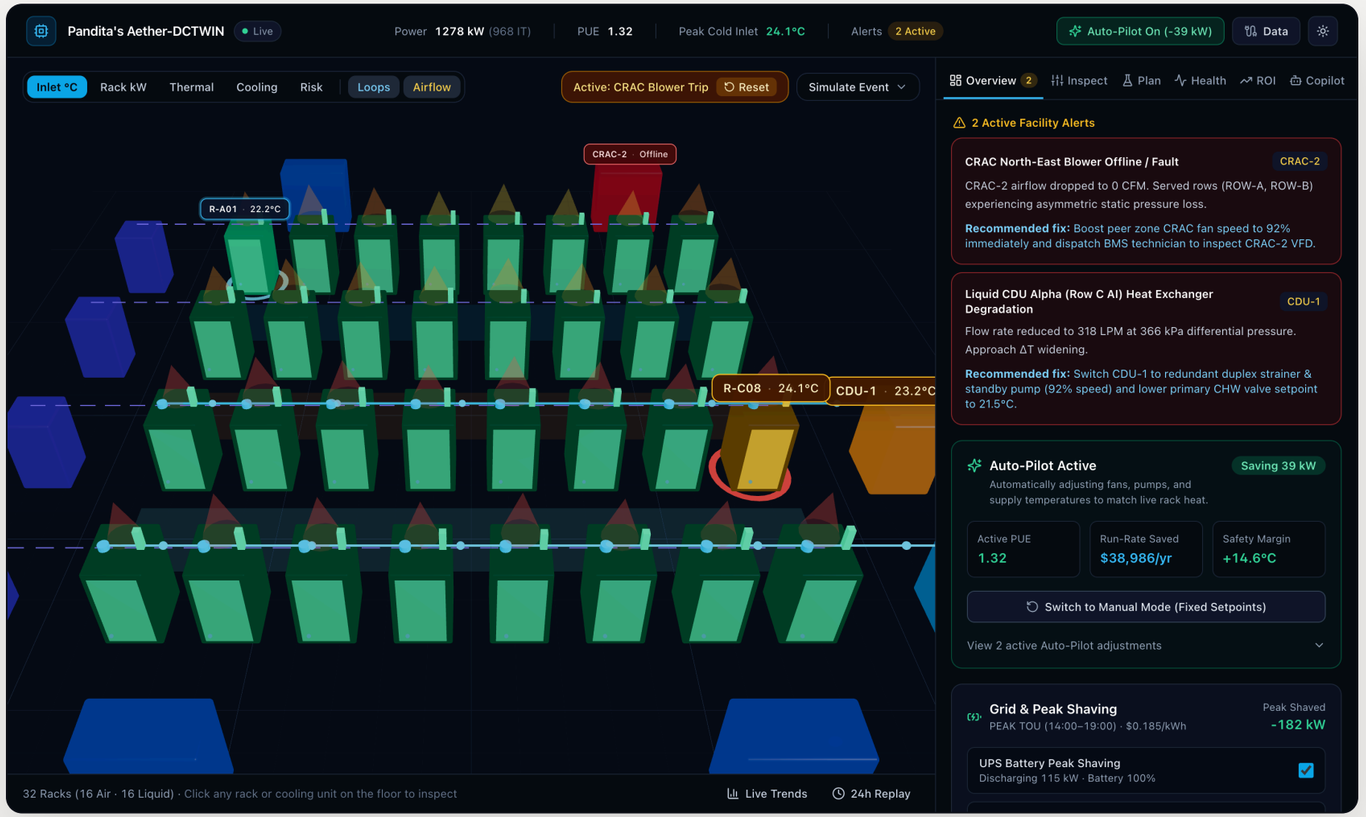
When a perimeter unit trips (CRAC-2 Blower Failure), the platform highlights the failed unit and affected Row B racks instead of flooding operators with 20 separate alerts.

Visualise airflow spillover & 42U chassis impact

Toggle live airflow plumes on the 3D floor and click any rack to inspect 42U server elevations, dual A/B PDU feeds and liquid manifold temperatures.

Spend time fixing, not searching

Plain-English fault cards explain likely cause, thermal runway and immediate mitigation steps.



LIVE PLATFORM • FAULT TRIAGE

CRAC-2 offline root-cause alert, airflow plumes and recommended fix

DECISION SUPPORT • GROUNDED FACILITY COPILOT

Plain-English answers backed by live meters

Give shift engineers and site leads instant answers across telemetry, procedures and capacity — without guesswork.

Every number comes from deterministic tools

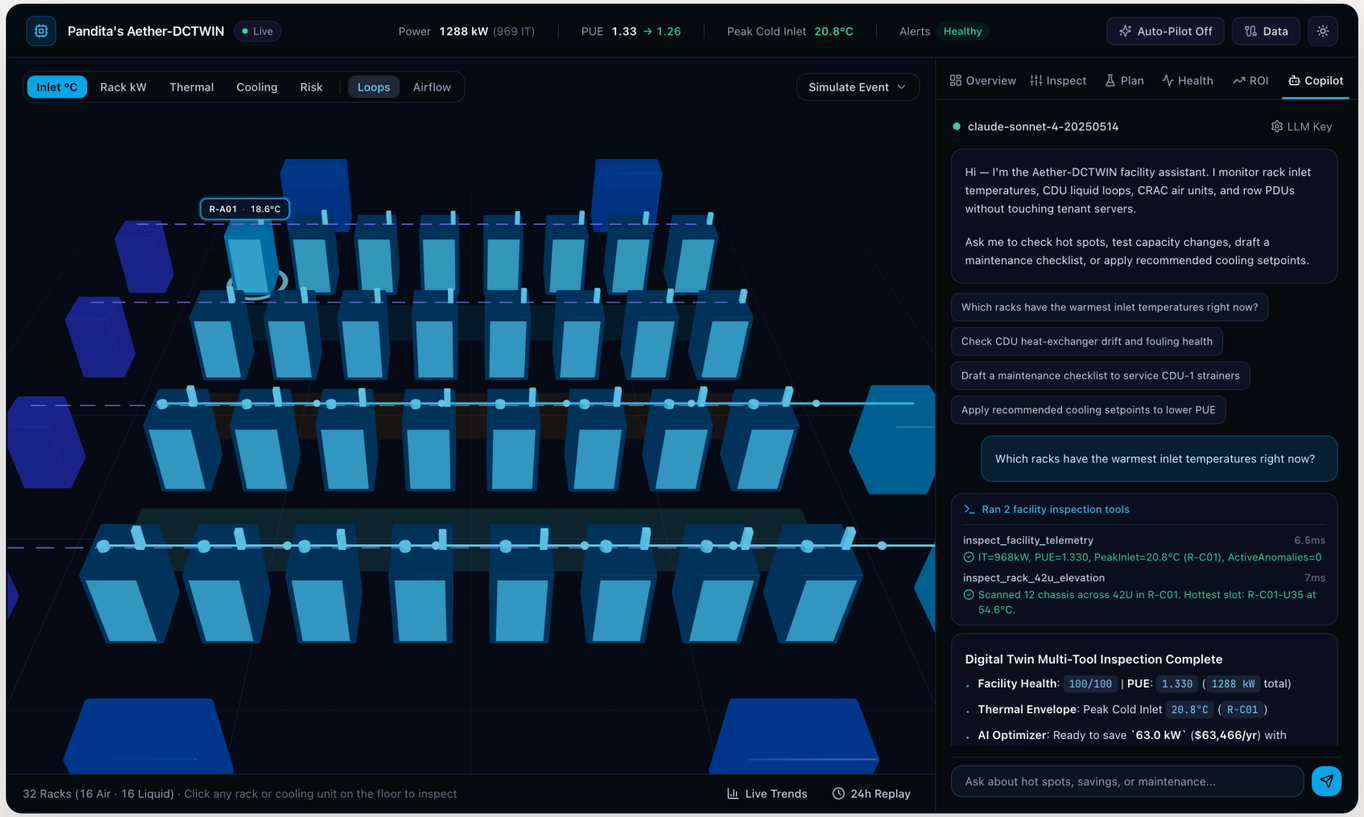
Rather than guessing, the assistant calls read-only facility inspection tools to query live rack sensors, CDU loops and historical trends.

Transparent audit trail on every reply

Engineers can expand the exact inspection tools executed on every question, keeping troubleshooting transparent and verifiable.

Flexible model choice (Cloud or On-Prem)

Supports enterprise LLM endpoints or air-gapped local deployment alongside a deterministic offline physics fallback.



LIVE PLATFORM • COPILOT WORKSPACE

Plain-English diagnostics with expandable facility inspection tool trace

MEASURED VALUE • INLINE ROI & 24H HISTORIAN

Prove financial return and replay every change

Translate operational improvements directly into annual cost savings, unlocked IT capacity and auditable history.

Live financial & sustainability case

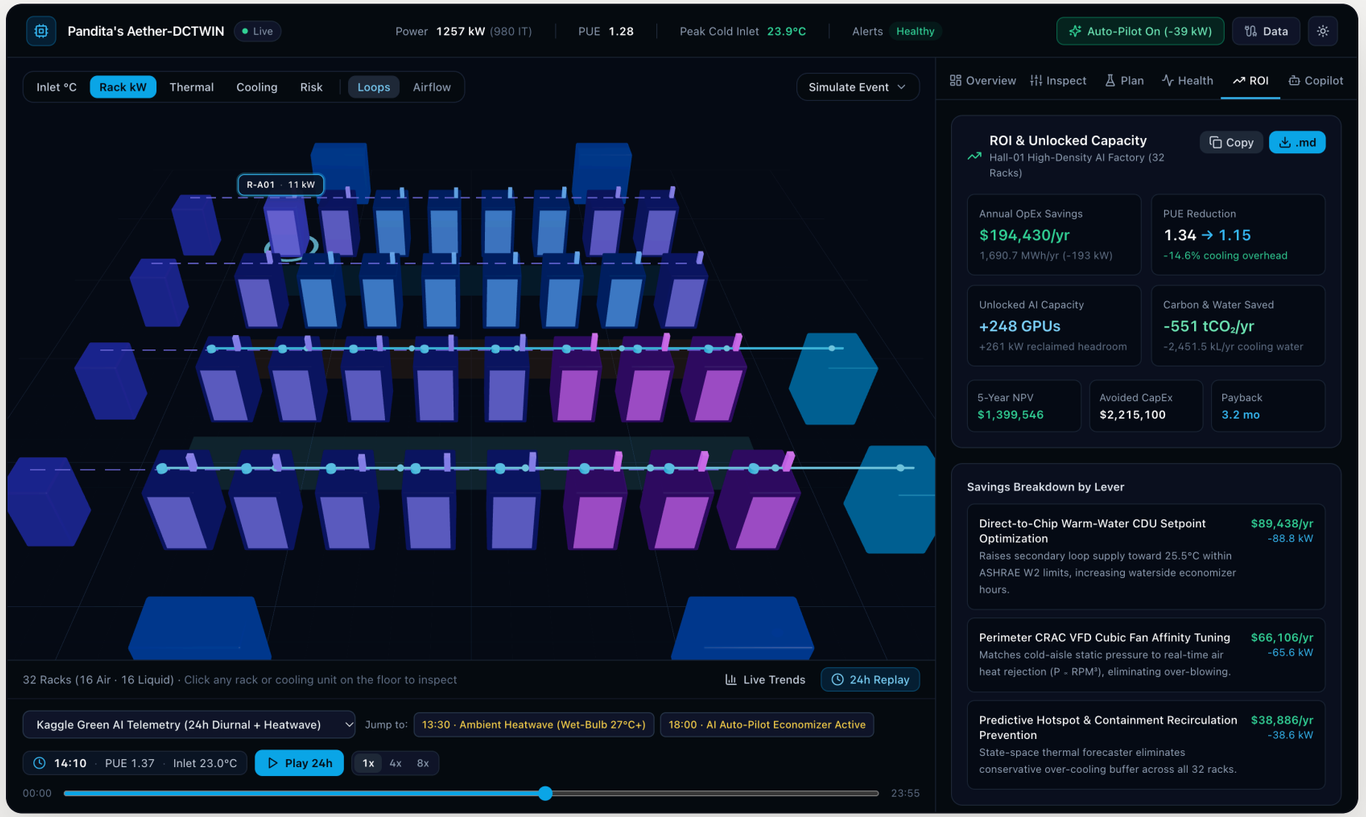
Dedicated ROI tab converts kW saved into annual OpEx savings, CO_2 reduction, payback period and 5-year net present value.

Quantify unlocked GPU capacity

Shows how much reclaimed cooling power can be reallocated to revenue-generating AI racks under the existing utility cap.

24-hour DVR timeline replay

Built-in DuckDB historian lets teams scrub backwards across the past 24 hours to review incidents or verify before/after trial gains.



LIVE PLATFORM • ROI & 24H REPLAY

Inline business case metrics paired with the bottom 24-hour historical scrubber

BEYOND THE DATA HALL • COMMERCIAL & BUSINESS TEAMS

Faster customer responses and less everyday admin

Connect site capacity and commercial documents so customer-facing and support teams move faster.

SALES & CAPACITY FIT

Qualify opportunities faster

Match customer density, power and delivery requirements against real hall capacity in minutes.

PROPOSALS & RFPS

Prepare technical responses

Draft accurate RFP answers from approved site specs, PUE data and past proposals.

MARKETING

Make outreach relevant

Build account briefs and campaigns grounded in verified site connectivity and sustainability metrics.

PROCUREMENT

Compare supplier offers

Check CDUs, switchgear and service quotes against technical specs, lead times and commercial terms.

FINANCE

Resolve billing gaps

Reconcile customer contracts against metered power usage and SLA commitments to prevent disputes.

HR & SITE READINESS

Get people site-ready

Coordinate technician and contractor onboarding, safety certifications and access approvals.

Typical systems: CRM • proposal library • DCIM • ERP • billing / metering • HRMS

HOW WE WORK

Start small. Prove the value.

Work alongside your existing BMS, DCIM and engineering teams — without disrupting live operations.

1 Choose one problem

Agree a focused goal — cooling savings, stranded capacity or early fouling detection — and ingest a CSV export or read-only meter feed.

2 Test in Shadow Mode

Calibrate the digital twin on your hall data and review live recommendations with your engineers before changing a single setpoint.

3 Expand what works

Measure the kW, PUE and operational gains, then roll out approved workflows across additional halls and sites.

Your team stays in control.

Read-only BACnet / Modbus / SNMP / CSV • Zero tenant agents • Instant manual fallback

NEXT STEP

Let’s choose a starting point

Bring your operations, engineering and business leads together for a live walkthrough and scoping session.

Where could AI make the biggest difference at your site?

What BMS, power meter or historical CSV data can we start with?

What would a clear 4-to-6 week pilot result look like?

We will walk through the live digital twin and propose a clear scope, expected benefit and test plan.